CLIENT-SIDE BIOPHOTONICS

Pulse from Pixels — Toss the Sensor

Camera-Based Photoplethysmography (PPG) Vital Signs Monitor

- Goal: Measure human heart rate accurately using only a smartphone camera and flashlight LED.
- Zero Wearables: Eliminates smartwatches, finger clips, and expensive pulse oximeter hardware.
- 100% Private in Browser: Video frames stay in transient memory and are discarded immediately. Zero cloud upload.
- Modern Tech Stack: Built with React 19, TypeScript, and causal hand-crafted DSP algorithms.

STAGE 01 / PRINCIPLE OF OPERATION

How Can a Camera See Your Heartbeat?

Photoplethysmography (PPG) & The Beer-Lambert Law

- Arterial Pulsation: Every heartbeat pumps blood into the microvascular capillary bed of your fingertip.
- Light Absorption: Hemoglobin in blood strongly absorbs green light (~540 nm wavelength).
- Microscopic Brightness Shifts: More blood in fingertip = darker frame. Less blood between beats = brighter frame.
- Optical Sensing: Phone camera records at 30 frames/second. Spatial averaging converts video frames into a 1D optical pulse wave.

STAGE 02 / MATHEMATICAL PIPELINE

Step-by-Step Processing Pipeline

From Raw Camera Pixels to Verified Heart Rate

- 1. Spatial Downsampling: Scales video to 64x64 ROI, extracting spatial mean to save battery and guarantee O(1) memory.
- 2. Uniform Resampling: Linear interpolation maps jittered mobile timestamps onto an exact 30.0 Hz time grid.
- 3. 4th-Order Butterworth Filter: Direct Form II SOS removes DC baseline drift, breathing sway, and high-frequency noise.
- 4. Dual Consensus Engine: Time-domain peak detector cross-checks frequency-domain FFT peak within 3.0 BPM tolerance.

STAGE 03 / NOISE REJECTION

Cleaning the Signal: Band-Pass Filter

Why We Need a 4th-Order Butterworth Filter

- The Noise Problem: Respiration creates slow 0.2 Hz baseline drift. Hand tremor and sensor noise add 10+ Hz spikes.
- Passband Window: 0.5 Hz to 4.0 Hz (corresponds to physiological range of 30 BPM to 240 BPM).
- Cascaded Biquad Sections: Two 2nd-order Direct Form II Transposed sections give steep 24 dB/octave rolloff.
- Maximally Flat: Zero passband ripple prevents false systolic peak distortion.

STAGE 04 / DUAL ESTIMATION

Dual-Engine Consensus Cross-Check

Preventing False Readings via Time & Frequency Agreement

- Method A (Time Domain): Inter-Beat Interval (IBI). Detects systolic wave peaks and calculates BPM = 60 / Delta_t.
- Method B (Frequency Domain): 4,096-point Hann-windowed FFT. Finds the dominant spectral power peak frequency f_peak.
- Consensus Rule: If |BPM_Time - BPM_Freq| <= 3.0 BPM -> Status: Verified Clean.
- Robustness: If hands shake, the two methods disagree (>3.0 BPM) and the system automatically prompts user to steady finger.

STAGE 05 / ARCHITECTURE & PRIVACY

System Architecture & Memory Discipline

Why Client-Side Matters for Medical Signal Privacy

- Zero Server Backend: Entire DSP chain runs locally in browser JavaScript thread and Web Worker. No backend server required.
- Strict O(1) Memory Rule: Raw video frames are discarded immediately after computing spatial mean. Zero memory leaks.
- Complete Data Privacy: No personal video, camera feed, or physiological data is ever transmitted over network.
- Universal Accessibility: Runs on any standard mobile phone or laptop browser via lightweight static web deployment.

STAGE 06 / USER EXPERIENCE

Landing Page & Presenter Easter Egg

Crumple Editorial Design + Secret Academic Inspection Lab

- Public Landing Page: Monochromatic Crumple editorial aesthetic with arterial red pulse disc and AHA resting categories.
- Direct Action: "Start Camera" launches capture and transitions smoothly into the monitoring workspace.
- Secret Presenter Trigger: Clicking "Developed by Krishnajith KS" 6 times unlocks the Teacher Evaluation Workbench.
- Inspection Tools: Multi-channel optical viewport, live waveform stack, viva diagnostics, and synthetic benchmarks (54, 72, 108 BPM).

STAGE 07 / VIVA DEFENSE CHEAT SHEET

Examiner Questions & Quick Answers

Core ECE Viva Concepts in Plain Language

- Q: Why green channel over red? A: Oxygenated hemoglobin absorbs green light (~540nm) 2.5x more strongly, giving superior SNR.
- Q: What if phone drops frames? A: Linear resampler maps irregular timestamps to an exact 30 Hz uniform grid so filter cutoffs do not drift.
- Q: Why use Direct Form II Biquads? A: Transposed Direct Form II minimizes floating-point roundoff error and state register overhead (flat O(1)).
- Q: Is this a clinical medical device? A: No, it is an educational biophotonic DSP monitor demonstrating optical plethysmography in browser.